

protection algorithm, having no apparent structure or period.

The baseline acquisition approach uses direct acquisition of the M code navigation signal, obtaining processing gain through the use of large correlator circuits in the user equipment. The M code signal has been designed for autonomous acquisition, so a receiver can acquire the M code signal without access to C/A code or Y code signals.

Misleading the location server

Typically, in a system, the location server (LS) is a secure user plane location (SUPL) platform (OMA-AD-SUPL). It may be a serving mobile location centre (SMLC) in a GSM network (3GPP TS 23.271), a standalone SMLC (SAS) in a UMTS network (3GPP TS 23.271) or another type of network node. The SUPL location platform (SLP) is a network entity on the Internet that is used to facilitate location. The user-plane location protocol (ULP) is an HTTP-based protocol, used between the SLP and the SUPL-enabled terminal (OMA-TS-ULP). The SLP has a connection to a global navigation satellite system (GNSS) reference server in order to retrieve and cache assistance data.

Location requests are initiated either from the SUPL-enabled terminal (SET), which is known as a SET-initiated transaction, or from the network, which is known as a network-initiated transaction. A network-initiated request is made by a location-based application (LBA), which sends a request to the SLP for the location of a particular handset. The SLP performs the messaging with the SET and determines the location before returning that location to the LBA.

When an A-GPS location fix is required, the SLP calculates the GPS assistance data that is specific to the approximate location of the SET. When the SET is in a cellular network, the approximate location generally comes from the coverage area of the serving

cell. The SET provides the identification of the cell (cell-ID) to the SLP. The SLP determines which satellites are in view from the approximate location and provides to the SET assistance data for those satellites.

The assistance data types sent to the SET depend on the mode of A-GPS. In handset-based A-GPS, the SLP generally provides the navigation model, ionosphere model, reference time and reference location. The handset uses this information to lock onto the satellites and calculate a location. It then returns the location to the SLP. In handset-assisted mode, the SLP

One source of the orbital data is the International GNSS Service (IGS). This is used to determine the location of the satellites for the given time. The spoofer calculates the range of each satellite in view of the desired location and uses that as the basis for determining the pseudo-range measurements. The measurements are generally converted into pseudo-ranges by simulating a clock error and introducing other errors such as the ionosphere, troposphere and other random errors.

The key piece of information that the spoofer needs to provide to the SLP

Spooing is more serious than jamming because jamming causes the service to degrade in performance, while spoofing takes control of the user receiver

provides the acquisition assistance and the reference time for the handset to lock onto the satellites and return the measurements to the SLP. The SLP invokes the position calculation function in order to calculate the location of the handset.

The spoofer must provide GPS measurements that result in the LS calculating a location that is desired by the spoofer (or attacker). For the network-initiated case, once the LS (SLP) determines the location of the handset, the location is provided to the network entity that requested it. If the LS is trusted by the recipient of the location, the location is considered to be valid even though it may not be.

The aim of the spoofer is to convince the LS to provide the location that the attacker desires. The attacker does this by falsifying (or spoofing) the measurement data such that the location provided by the LS is effectively predetermined by the attacker.

In order to spoof his location, the spoofer needs the satellite ephemeris. The ephemeris may be from a request to the SLP for assistance data. Alternatively, the ephemeris may come from another source such as the Internet.

is the cell-ID. From that, the SLP will look up the coverage area of the cell and calculate the assistance data. This coverage area will also use that as the initial location estimate for the position calculation function (PCF). The cell-ID is also often used for location assurance on the SLP. For the spoofer, one way of knowing the cell-IDs is to log cell-IDs against locations and build up a database over time.

Countermeasures to location server spoofing

This kind of spoofing can also be detected by the server. A server with anti-spoofing capability will be able to detect the measurements being spoofed. Some of the obvious signs on the server of spoofing are:

1. Clock error: The receiver clock error is very small—less than 1×10^{-10} second.
2. Residuals: The residuals calculated as part of the least squares process are very small.
3. Uncertainty ellipse: The uncertainty ellipse is very small (less than one metre of uncertainty).

A more clever spoofer will manipulate the measurements by introducing